

**DASM II VDB
Installation Manual**

July 1999

PN 39-400077 R00

DASM II VDB Installation Manual

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*The World Resource
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Part Number

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LICENSE SECTION

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Safety Precautions

Cautions Symbols

The following symbols appear in this manual:



CAUTION: Hazardous Voltages present. Reduce the risk of electric shock and danger to personal health by following instructions.



CAUTION: Advises against actions that could damage equipment, produce inaccurate data, or invalidate a procedure.

Mesures de sécurité

Symboles

Vous trouverez ci-dessous la signification des différents symboles utilisés:



ATTENTION: présence de tensions dangereuses. Pour éviter les risques d'électrocution et de danger pour la santé physique, veuillez suivre les instructions.



ATTENTION: risques de blessures corporelles et de dégâts matériels. Veuillez suivre les instructions.

Sicherheitsvorkehrungen

Symbole

Die Symbole in diesem Handbuch haben folgende Bedeutung.



ACHTUNG: Gefährliche Spannungen. Anweisungen befolgen, um Stromschläge und Verletzungen zu vermeiden.



ACHTUNG: weist auf Handlungen hin, die das Gerät beschädigen könnten, unrichtige Daten verursachen oder einen Vorgang auslösen könnten.



CAUTION: SHOCK HAZARD – Due to the risk of electrical shock, this instrument must be installed only by qualified personnel.

ATTENTION: RISQUE DE CHOC ÉLECTRIQUE - Risque de choc électrique. Cet instrument doit être installé SEULEMENT par une personne qualifiée.

ACHTUNG: Elektroschockgefahr - Wegen Elektroschockgefahr darf dieses Instrument nur von Fachleuten installiert werden.



CAUTION: The integrated circuits in the DASM II VDB are sensitive to electrostatic discharge (ESD). Be sure to follow proper procedures for handling, replacing, storing and transporting ESD-sensitive components.

ATTENTION: Les circuits intégrés du DASM II VDB sont sensibles aux décharges électrostatiques (ESD). S'assurer de suivre les bonnes méthodes de manutention, de remplacement, d'entreposage et de transport des composants sensibles aux perturbations des décharges électrostatiques (ESD).

ACHTUNG: Die eingebauten Schaltkreise in den DASM II VDB sind auf Elektrostatik (ESD) empfindlich. Vorschriftsmäßige Handhabung, Auswechslung, Lagerung und Transport von Elektrostatik-empfindlichen Komponenten befolgen.



CAUTION: Be sure the DASM II VDB is unplugged before attempting to access the fuse.

ATTENTION: Vérifiez que le DASM II VDB est débranché avant de tenter d'accéder au fusible.

ACHTUNG: Feststellen, daß der DASM II VDB Stecker herausgezogen ist, bevor an die Sicherungen herangegangen wird.



CAUTION: You must use a fuse of the correct type and rating for the currently configured line voltage.

ATTENTION: Il faut utiliser un fusible de la capacité et du type appropriés pour la tension secteur configurée actuellement.

ACHTUNG: Typ und Belastbarkeit der verwendeten Sicherung müssen für die eingestellte Netzspannung angezeigt sein.

The following general safety precautions must be observed during all phases of service and repair of this equipment. Failure to comply with these precautions or with specific warnings elsewhere in this manual violates safety standards of design, manufacture, and intended use of this equipment.

Do Not Operate In An Explosive Atmosphere

Do not operate the equipment in the presence of flammable gases or fumes. Operation of any electrical equipment in such an environment constitutes a definite safety hazard.

Keep Away From Live Circuits

Operating personnel must not remove equipment covers. Only Qualified Service Personnel may remove equipment covers for internal subassembly or component replacement or any internal adjustment. Do not replace components with power cable connected. To avoid injuries, always disconnect power and discharge circuits before touching them.

Do Not Substitute Parts Or Modify Equipment

Because of the danger of introducing additional hazards, do not install substitute parts or perform any unauthorized modification of the equipment.

Power Cord Connection



CAUTION: DASM II VDB is designed to work with single-phase power systems having a grounded neutral conductor. To reduce the risk of electric shock, do not plug DASM II VDB products into any other type of power system. Contact your facilities manager or a qualified electrician if you are not sure what type of power is supplied to your building.

ATTENTION: DASM II VDB sont conçus pour fonctionner avec des alimentations monophasées munies d'un conducteur neutre mis à la terre. Pour écarter les risques d'électrocution, ne pas brancher de DASM II VDB un autre type d'alimentation secteur. En cas de doute quant au type d'alimentation électrique du local, veuillez vous adresser au directeur de l'exploitation ou à un électricien qualifié.

ACHTUNG: DASM II VDB sind für den Betrieb an Einphasen-Stromnetzen mit geerdetem Nulleiter vorgesehen. Um die Stromschlaggefahr zu reduzieren, schieß Sie DASM II VDB nicht andere Stromquellen an. Ihr Betriebsleiter oder ein qualifizierter Elektriker kann Ihnen die Daten zur Stromversorgung in Ihrem Gebäude geben.

REGULATORY COMPLIANCE

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

Shielded cables must be used with this unit to ensure compliance with the Class A FCC limits.

EC Declaration of Conformity

We, Analogic Corporation
Centennial Industrial Park
8 Centennial Drive
Peabody, Ma. 01960

declare under sole responsibility that the: DASM II VDB Data Acquisition System Manager, to which this declaration relates, meets the essential health and safety requirements and is in conformity with the relevant EC Directives listed below using the relevant section of the following EC standards and other normative documents;

EC EMC Directive 89/336/EEC

Essential health and safety requirements relating to electromagnetic compatibility.

EN 55011:1991 + AM1:1997 and Am2:1996 Group 1, Class A
Limits and methods of measurement of radio disturbance characteristics of information technology equipment.

IEC 801-2:1991 Electrostatic discharge requirements for industrial process, measurement and control equipment.

IEC 801-3:Draft Radiated electromagnetic field requirements for industrial process, measurement and control equipment.

IEC 801-4:1988 Electrically fast transients for industrial process, measurement and control equipment.

IEC 801-5:Draft Testing and measurement techniques - Section 5: Surge Immunity.

EN60601-1-2 Collateral Standard: Electromagnetic compatibility - Requirements and tests.

EC Low Voltage Directive 72/23/EEC

Essential health and safety requirements relating to electrical equipment designed for use with certain voltage limits.

EN 60950 (1992), A1+A2+A3+A4+A11 Safety of information technology equipment including electrical business equipment.

The safety issues of this information technology equipment type have been evaluated by a government accredited European third party organization such as ETL ID No. 0344.

EC Medical Device Directive 93/42/EEC Annex II Section 3.

Essential health and safety requirements Annex (I) relating to electrical medical equipment (Medical Devices). Conformity Assessment by TÜV Rheinland ID No. 0197, Annex II Certificate No. HD 9910023.

The CE mark has been affixed on the device according to Article 17, of the EC Medical Device Directive 93/42/EEC.

Date of Manufacture: Refer to Nameplate

Serial Number: Refer to Nameplate



A handwritten signature in cursive script, which appears to read 'Robert H. French'.

Vice President QA & RA

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Introduction

PURPOSE OF THIS MANUAL

The DASM II VDB Installation Manual describes how to set up and test the Data Acquisition System Manager (DASM) II Video Display Buffer (VDB). It includes information on hardware installation, connecting cables, startup, shutdown, troubleshooting, and other related topics.

AUDIENCE

This manual is intended for any person responsible for installing or maintaining the DASM II VDB.

DOCUMENT STRUCTURE

The DASM II VDB Installation Manual contains three chapters and four appendices, as follows:

Chapter 1	Presents an Overview of the DASM II VDB
Chapter 2	Describes installation and start-up including cabling information
Chapter 3	Describes configuring the DASM II VDB
Appendix A	Describes General Safety Precautions and provides safety specifications
Appendix B	Presents Troubleshooting Information
Appendix C	Presents Technical Diagrams and Functional Descriptions
Appendix D	Lists DASM II VDB Specifications

Chapter 1

Overview

1.1 **PRODUCT DESCRIPTION**

The DASM II VDB provides a filming interface between a host computer and a laser imager compliant with the '952' protocol. It receives digital data from the host computer via a SCSI port and produces a video output identical to the Genesis IC/OC.

The DASM II VDB connects to the host's SCSI port and the laser camera's data ports, as shown in Figure 1-1 on the following page.

Circuitry is contained on one printed circuit board assembly. The functionality of this circuitry may be broken into two logical blocks; the Data Acquisition System Manager (DASM) and the Video Display Buffer (VDB).

The DASM II VDB accepts image data and commands from a host via a SCSI bus. The DASM processes the commands, and transfers the image data to the VDB. Commands and status are transferred to the camera over an RS-422 serial line. All image data is presented to the laser imager as composite video.

The entire system is packaged in a stand-alone enclosure containing a printed circuit board, a power supply, and all necessary cables.



CAUTION: SHOCK HAZARD – Due to the risk of electrical shock, this instrument must be installed only by qualified personnel.

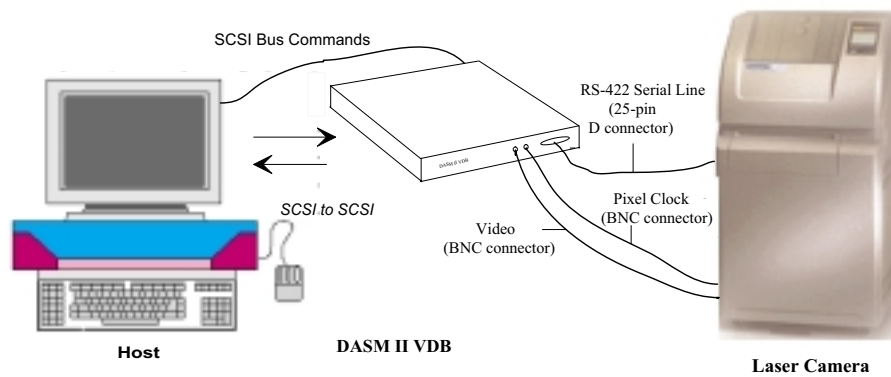
1.2 DASM II VDB FUNCTIONAL OVERVIEW

The DASM II VDB performs the following tasks:

- ❑ Accepts digital images from the host, over the SCSI bus
- ❑ Converts digital images to analog
- ❑ Communicates with the laser Camera using the 952 protocol

The figure below shows the DASM II VDB in a standard operating environment:

Figure 1-1: DASM II VDB Functional Flow



Chapter 2

Installation and Start-Up

2.1 OVERVIEW

This chapter explains how to connect the cables and start the DASM II VDB.

The DASM II VDB comes in a standard configuration. If you plan to change the configuration, see Chapter 3.

2.2 UNPACKING THE UNIT

The DASM II VDB consists of the following items:

- ☐ DASM II VDB
- ☐ SCSI Cable
- ☐ Power Cord

Note that the customer must provide the following items:

- ☐ Video Cable
- ☐ Pixel Clock Cable (if required)
- ☐ Serial Host Interface Cable

Verify that you have everything.

2.3 CABLING INFORMATION

The DASM II VDB connects to the host's SCSI port with a 50-pin SCSI Compliant cable. Also, there are a minimum of two cable connections required between the DASM II VDB and the camera: a shielded RS-422 serial-line cable for serial communications and a 75-ohm coaxial cable to carry the Video signal. Depending on the installation, a second 75-ohm coaxial cable may be needed to provide the Pixel Clock to the laser camera.

2.3.1 SCSI Connections

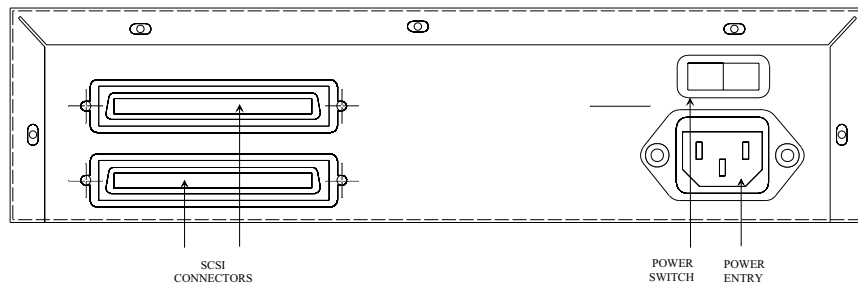
The figure below shows the SCSI connectors at the rear of the DASM II VDB enclosure.

The following symbols appear in this manual:



CAUTION: All units must be powered-down before connecting the interface cables.

Figure 2-1: DASM II VDB Rear Panel



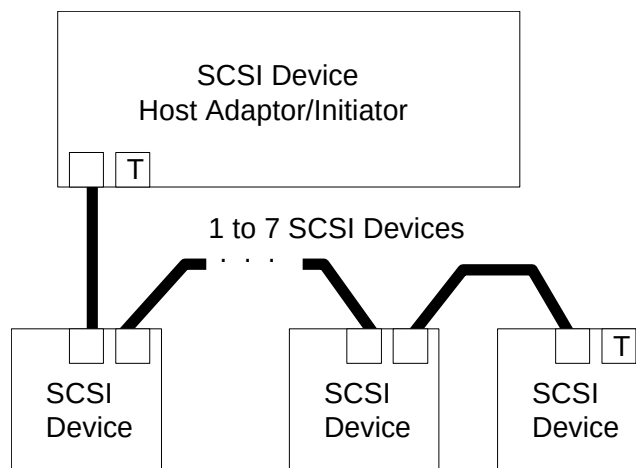
The SCSI cables must be connected, with the strain relief clips locked in place, as follows:

- ❑ The SCSI cable between the host and DASM connects to one of the DASM's SCSI ports.
- ❑ The other SCSI port of the DASM II VDB must be either terminated **OR** connected to the next daisy-chained SCSI device. See the following section, *Connecting Other SCSI Devices*, for more information on daisy-chaining SCSI devices.

2.3.2 Connecting Other SCSI Devices

The DASM II VDB can be "daisy-chained" with other SCSI devices. Up to 8 SCSI devices (including the host) may be daisy-chained together, as illustrated in the figure below:

Figure 2-2: Daisy-Chaining SCSI Devices



T Indicates SCSI Termination

The SCSI devices at both ends of the cable must be terminated.

- ❑ When DASM II VDB is not the last device in the chain the SCSI terminator must be removed from the DASM II VDB's SCSI Connector **AND** the Active SCSI Termination must be turned OFF.
- ❑ Otherwise, when the DASM II VDB is the last device in the chain, a SCSI terminator must be connected **OR** the Active SCSI Termination Switch must be set ON.

Chapter 3 shows the location of the Active Termination Switch.

Note: The length of all the SCSI cables in a chain must be less than 6.0 meters.

2.3.3 Laser Camera Control

The 25-pin 'D' Connector on the front panel of the DASM II VDB provides the RS422 Serial Line used to control the Laser Camera.

Pin assignments for the DASM II VDB's Camera Control Connector is located in Appendix D: *Specifications*.

2.3.4 Video and Pixel Clock

There are two BNC Connectors on the DASM II VDB front panel labeled as follows:

- ❑ Video Output – This connector provides the image in composite video format. Timing and waveform details are located in Appendix D.
- ❑ Pixel Clock Output – Some laser camera might require an externally supplied pixel clock. The Pixel Clock Output is driven by TTL. Details are listed in Appendix D.

2.4 START-UP

This section describes the start-up sequence, as well as how to recognize error conditions.

2.4.1 Powering On

This following sequence occurs on the front panel's LEDs when the power switch is turned on:

1. The power LED lights.
2. The octal SCSI ID flashes briefly. The Least Significant Bit is next to the power LED. Note that if the SCSI ID is zero, no LEDs flash.
3. These LEDs flash once and go out.

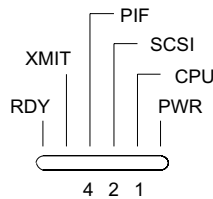
Once the power-up sequence completes, the DASM II VDB CPU LED (second light from right) blinks regularly, indicating that the CPU is active and functioning normally.

2.4.2 DASM II VDB LED Indicators

The front panel of the DASM II VDB provides a visible indication of operational status in the form of six green LEDs. Of these, four are specific to the DASM Controller and two are specific to the Video Display Buffer.

The DASM II VDB front panel is illustrated below:

Figure 2-3: Status Indicators



From left to right, the indicator lights on the DASM II VDB perform the following functions:

- ❑ RDY - Image Ready light indicates that an image has been transferred, and is stable at the VDB video output.
- ❑ XMIT - Image Transmit light indicates that an image is being transferred from the Common Memory to the VDB Field Memory. Note that a printable video image is not available during an image transfer.
- ❑ PIF - Personality Interface light activity, in the VDB application, coincides with the image transmit function.
- ❑ SCSI - This light indicates the host is accessing the DASM via the SCSI bus, such as commands sent or data received.
- ❑ CPU - This light indicates that the DASM CPU is active and operating properly. It flashes continuously ("blinks" on and off) following power-up.
- ❑ PWR - The Power Light indicates that the DASM is powered. If this light does not illuminate at power-up, you should first check the power cord connections.

2.4.3 *Start-Up Sequence*

When the DASM II VDB is powering up, it performs start-up diagnostics; and other related actions.

At power-up, the DASM performs the following tests:

- ☐ EPROM checksum test
- ☐ MFP 68901 access test (the Multi-Function Peripheral chip controls the serial port, timing, and I/O)
- ☐ SCSI register access test
- ☐ Static RAM test
- ☐ Dynamic RAM test
- ☐ DMA test

When the DASM CPU LED blinks continuously at about two flashes per second, the initialization sequence is complete.

If an error occurs, the error condition is reported by CPU, SCSI, and PIF LEDs blinking in tandem followed by an error code.

2.4.4 *Self-Test Continuous Mode*

Continuous Mode is the default setting. After executing the start-up diagnostic, the resident software will continuously run the VDB Self-Test. During Self-Test images are constructed in Data Memory, then transferred to the Field Memory via a PIF DMA. Self-Test will continue until the first SCSI command is received by the DASM.

Chapter 3

Configuring the DASM II VDB

3.1 OVERVIEW

This chapter describes how to configure the DASM II VDB.

3.2 USER SETTINGS

The DASM II VDB is initially set up as follows:

Accessible Externally

1. SCSI ID set to zero
2. SCSI Termination Switch set to OFF position

Setting Requiring Access to the Printed Circuit Assembly

1. RAM diagnostic disabled
2. Manufacturing diagnostics disabled
3. Self Test-Loop enabled
4. Video Frame Rate set to 60 Hz
5. Safety Ground open

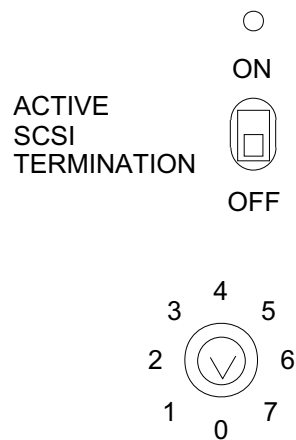
3.2.1 SCSI ID and SCSI Termination

The SCSI ID Switch is accessible from the bottom of the DASM II VDB enclosure. With a small flat blade screwdriver simply rotate the switch shaft to the desired setting.

Setting the Active SCSI Termination switch to ON enables the internal SCSI termination. If you use this method to terminate, do not install an external terminator in the SCSI port in the rear of the DASM II VDB enclosure.

If lit, the LED above the Active Termination switch indicates Active SCSI Termination is turned ON, and that termination power is being supplied.

Figure 3-1: DASM II VDB Bottom



3.3 VIDEO FORMAT AND SAFETY GROUND



CAUTION: SHOCK HAZARD – Due to the risk of electrical shock, this instrument must be installed only by qualified personnel.

To set the video format and safety ground settings it will be necessary to open the DASM II VDB enclosure and access the printed circuit board. See Figure 3-3 for the locations of the Video Format switch and the Safety Ground jumper.

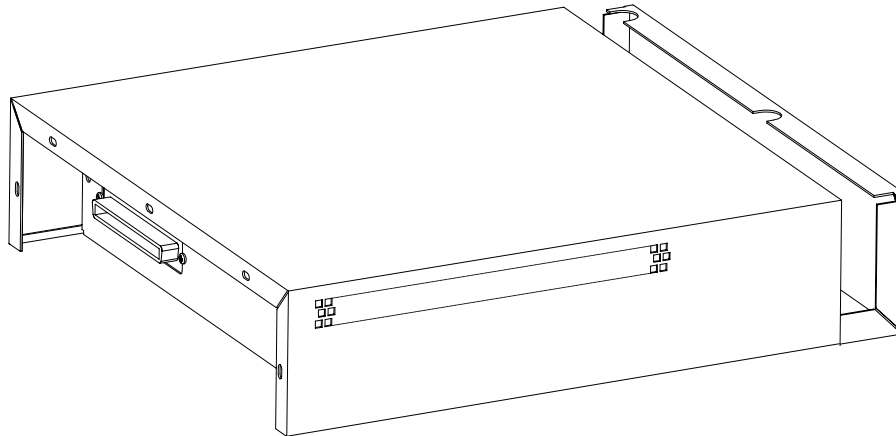
3.3.1 Removing the DASM II VDB COVER

To select Self-Test options it will be necessary to open the DASM II VDB enclosure and access the printed circuit board. To remove the DASM II VDB enclosure cover:

1. Perform shutdown and power off the DASM.
2. Unplug the power cable from the rear of the DASM.
3. Using the Phillips head screwdriver, remove the five screws on the back of the DASM enclosure. These five screws secure the system enclosure cover. See Figure 3-2 below.

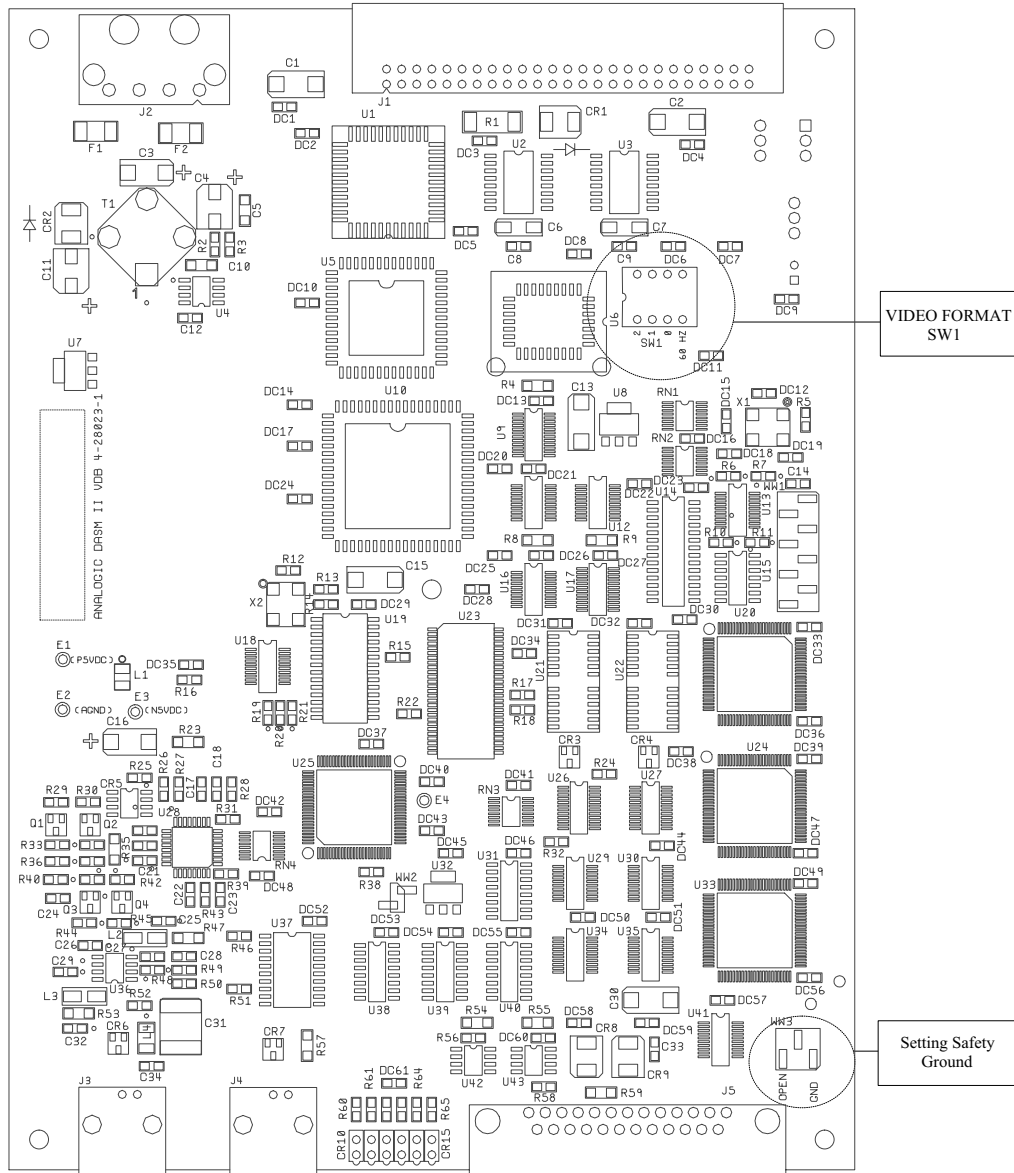
To remove the enclosure cover, pull it to the rear for approximately $\frac{1}{2}$ to $\frac{3}{4}$ inches, so that the tabs in the cover are free from the chassis bends making up the front panel. Then lift it up and away from the DASM enclosure.

Figure 3-2: DASM II VDB Enclosure



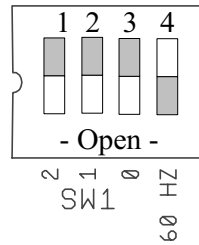
3.3.2 DASM II VDB Board

Figure 3-3: DASM II VDB Board



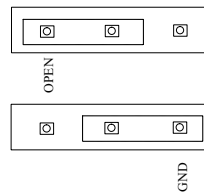
3.3.3 Setting the Video Format

When SW 1 position marked “60 Hz” is open (the default) the 60 Hz video format is selected. Closing this position selects the 50 Hz format. Refer to Figure 3-3 for the location of SW1.



3.3.4 Setting Safety Ground

Installing a shunt across the pins of WW3 marked GND connects pin 1 of the 25-pin Camera Control Connector to the Chassis Ground (Safety Ground). Otherwise, pin 1 is OPEN. Refer to Figure 3-2 for the location of Safety Ground.



Appendix A

Safety

A.1 INTRODUCTION

This appendix points out standard precautions to be observed while working with the DASM II VDB.

A.2 EQUIPMENT PRECAUTIONS



CAUTION: SHOCK HAZARD – Due to the risk of electrical shock, this instrument must be installed only by qualified personnel.

Properly power-down the DASM II VDB and disconnect the line cord from its power entry connector before accessing any internal components. Only qualified repairmen should open the DASM II VDB enclosure.

A.3 POWER REQUIREMENTS

Input range: 100-240VAC
Frequency: 50-60Hz
Current: 0.6A

A.4 POWER SUPPLY

A.4.1 Safety Warning

The power supply manufacturer gives the following warning; “Beware of internal lethal voltages due to charged capacitors even after AC input is disconnected. A minimum of 5 minutes should be allowed after AC Input power is disconnected before handling.”

A.4.2 Power Supply Fuse (Chart)



CAUTION: Be sure the DASM II VDB is unplugged before attempting to access the fuse.



CAUTION: You must use a fuse of the correct type and rating for the currently configured line voltage.

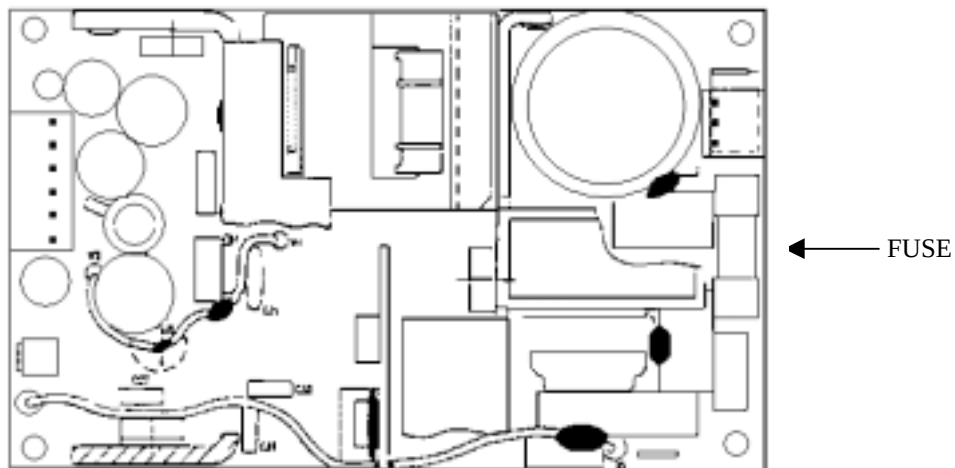
Listed below are the only two approved power supply replacement fuses.

Table A-1: FUSE Specification table

Analogic Part Number	Vendor P/N	Vendor Name
15-500193	21602.5	LITTLEFUSE INC.
15-500193	0001.1008	SCHURTER

A.4.3 Power Supply Layout

The power supply fuse is located as indicated below.



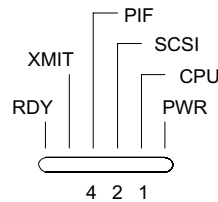
Appendix B

Troubleshooting

B.1 DASM II VDB LED INDICATORS

The DASM II VDB front panel has six LEDs, which are used for status and error information. These lights are illustrated below:

Figure B-1: Status Indicators



From left to right, the lights indicate the following:

- ❑ RDY - Image Ready light indicates that an image has been transferred, and is stable at the VDB video output.
- ❑ XMIT - Image Transmit light indicates that an image is being transferred from the Common Memory to the VDB Field Memory. Note that a printable video image is not available during an image transfer.
- ❑ PIF - Personality Interface light activity, in the VDB application, coincides with the image transmit function.
- ❑ SCSI - This light indicates the host is accessing the DASM via the SCSI bus, such as commands sent or data received.
- ❑ CPU - This light indicates that the DASM CPU is active and operating properly. It flashes continuously ("blinks" on and off) following power-up.

- ❑ PWR - Power Light indicates that the DASM is powered. If this light does not illuminate at power-up, you should first check the power cord connections.

B.2 START-UP SEQUENCE

When the DASM II VDB is powering up, it performs start-up diagnostics and other related actions.

At power-up, the DASM Controller performs the following tests:

- ❑ EPROM checksum test
- ❑ MFP 68901 access test (the Multi-Function Peripheral chip controls the serial port, timing, and I/O)
- ❑ SCSI register access test
- ❑ Static RAM test
- ❑ Dynamic RAM test
- ❑ DMA test

When the DASM CPU LED blinks continuously at about two flashes per second, the initialization sequence is complete.

If an error occurs, the error condition is reported by the CPU, SCSI, and PIF LEDs blinking in tandem followed by an error code.

B.2.1 Powering On

This following sequence occurs on the front panel's LEDs when the power switch is turned on:

1. The power LED lights.
2. The octal SCSI ID flashes briefly. The Least Significant Bit is next to the power LED. Note that if the SCSI ID is zero, no LEDs flash.
3. These LEDs flash once and go out.

Once the power-up sequence completes, the DASM CPU LED (second light from right) blinks regularly, indicating that the CPU is active and functioning normally.

B.3 SELF-TEST OPTIONS

Once the start-up sequence is complete, the resident software reads the state of the DIP switch labeled SW1 (see Figure 3-3 DASM II VDB Board) to determine whether further tests are enabled. To access SW1 you must first remove the top cover. Upon locating SW1 you will notice the numbers 0, 1, and 2 silk screened on the printed circuit board. The way Self-Test executes is determined by the setting shown below:

Switch Position	Continuous Mode	Serial Loop-back	Single Pass	Single Pass W/Memory	Debug Mode
0	Closed	Closed	Open	Open	Open
1	Closed	Open	Closed	Closed	Open
2	Closed	Closed	Closed	Open	Open

Continuous Mode is the default setting. After executing the start-up diagnostic, the resident software will continuously run the VDB Self-Test. Self-test will continue until the first SCSI command is received by the DASM.

Serial Loop Test: In this mode the VDB tests the RS422 Driver and Receiver. Prior to executing this a special loop-back connector must be plugged into the Camera Control connector on the front panel of the VDB, without the loop-back connector the test will fail.

The loop-back connector is made by simply adding two wires to a 25-pin Male 'D' connector: (pin 8 to pin 9) and (Pin 21 to pin 22).

Single Pass: The start-up diagnostics execute once, as described in B.2 Start-Up Sequence. Please be aware that when position 0 is open error and other messages encountered during start-up diagnostics are directed to the serial port. Otherwise the errors are encoded on the LEDs.

Single Pass W/Memory is controlled by Bit 2. When open the DASM exercises the entire DRAM during the start-up diagnostics. The Single Pass W/Memory test can take several minutes to complete, this is why it is normally disabled.

Debug Mode: When position 1 is open the program jumps to the I/O Monitor when a diagnostic error occurs. The I/O Monitor gives the basic tools to debug the hardware. Opening positions 0 and 2 allows the technician access to the start-up diagnostic suite via the serial port.

B.4 START-UP PROBLEM

To report diagnostic conditions, the LEDs on the DASM II VDB front panel have a corresponding binary representation. The figure below shows the value assigned to each LED. The section after the figure explains how the LEDs represent error codes. Note that this sections deals with error codes related to the DASM Controller, where Section B.5 explains the method used to report an error from the VDB diagnostic.

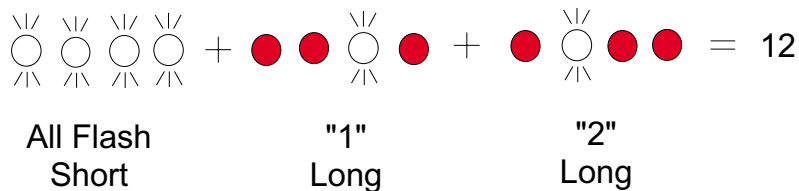
B.4.1 DASM Controller LED Error Codes

Using LEDs 2 and 4, a binary code signals any of the conditions listed in Table B-1 on the following page. The power light and PIF LEDs are not used to report diagnostic conditions.

The three LEDs permit a combined total of only seven error codes. To overcome this limitation, the codes are defined as a series of one or more patterns. Each pattern begins with all LEDs flashing briefly, to indicate the start of the number sequence that follows.

For example, to display error code 12 (Failed SCSI Interface Test), the CPU LED displays the first digit (1), stopping briefly before the SCSI LED displays the second digit (2), as shown in the figure which follows.

Figure B-2: Error Code Indicators



The display of each digit lasts approximately four times the duration of the initial binary "7" (that is, the flash of all LEDs). The pattern then repeats. Only the significant digits for each error condition are listed in Table 2-1, below:

Note: The first error stops the start-up sequence immediately.

B.4.2 Start-Up Error Codes

Table B-1: DASM Controller Start-Up Error Conditions

Value	Significance
1	Failed To Set Timer
2	Failed To Set Baud Rate
3	Failed To Access Receiver Status Reg. For Serial I/O
4	Failed To Start Refresh Clock
5	Failed In Set Up Of Serial I/O
6	Checksum Failed
7	Failed Static RAM Test
11	Failed I/O RAM Test
12	Failed SCSI Interface Test
13	Failed To Start VRTX Operating System
21	Bus Error
22	Address Error
23	Illegal Instruction
24	Undefined MFP (MC68901) Interrupt
25	Zero Divide Through Trace Trap Level 5
26	Chk, Trapv, Privilege, Or Trace Interrupt
31	Unknown Interrupt

B.5 VDB ERROR CODES

The following sequence is used to signal an error during the VDB portion of the Self-Test.

First, all three LEDs blink together for the duration of a single pass. This requires approximately 10 - 15 seconds. Then the specific error code blinks twice before the next pass begins, with all three LEDs blinking together as before.

The bit pattern of the error codes appears as a series of LEDs being turned on and off. The low order bit is first.

When two LEDs blink together, the bit value is 1.

When only one LED blinks, the bit value is 0.

To see an example of an error code displayed in this way, do the following:

- ❑ Remove the loop-back connector from the front panel of the DASM II VDB
- ❑ SW 1 position 0 is closed
- ❑ SW 1 position 1 is open
- ❑ Now cycle the AC power on the DASM
- ❑ When the first pass of the DASM Self-Test has completed, the LED display begins. The expected error is 5 (RS-422 port fail), and the LED sequence is:

Least significant first:

bit 0 = 2 LEDs on = 1 value 1

bit 1 = 1 LED on = 0

bit 2 = 2 LEDs on = 1 value 4

bit 3 = 1 LED on = 0

Table B-2: Serial Error Codes in LEDs

SERIAL ERROR CODES IN LEDs:

LED sequence, from left to right:

Checksum	1 2 1 1	checksum of EPROM
Serial	2 1 2 1	RS422 test - needs loop-back connector
DMA_setup	1 1 1 2	Tests DASM Block
Image_verify	2 1 1 2	tests full image transfer
DASM Block	2 2 2 2	nonspecific failure. SCSI to host is needed

B.6 RUN-TIME ERROR CODES

These codes may be found in the DASM Response Block byte 128, when the error bit (bit 2, where bit 0 is first) is set in Response Block byte 0. Only the first of multiple errors is stored in Response Block byte 128. Reading the Response Block clears the error. Note that the status "invalid command" is returned to the user directly in bit 0 of the SCSI status byte; it is not logged in the Response Block.

Table B-3: Run Time Error Codes

DASM Run-Time Error Codes

Error	Hex	Meaning
— General Error Codes —		
E_PARAM	81	Bad Parameter To Valid Command
E_NO_FF	82	No 0xFF Terminator In Command
E_TMO	83	Timeout
E_PIF_SELF	84	Peripheral Interface Self-Test Error
E_PIFNOTRDY	85	Peripheral Device Not Ready
E_OVRRUN	86	Data Overrun
E_UNDRUN	87	Data Under run
E_COM_LINK	88	Communications Link Error between DASM and PIF
E_DRAM	8A	Error in I/O RAM
E_EPROM	8B	EPROM Checksum Error

E_MFP	8C	Error in the MC68901 I/O and Timer Chip
E_SBIC	8D	Error in Access to SCSI Bus Interface Chip
E_NOTIMER	8E	No Timer Available (All In Use)
E_INTERN	8F	Internal System Software Error
E_INV_MSG	B0	Invalid/unexpected message from device
E_BAD_S	B1	Invalid S-record (from load routine)
E_ODD_ADR	B2	Odd address passed to function
E_BOUNDS	B3	I/O request overlapped buffer boundary
E_DIRECTION	B4	I/O request direction is invalid
E_NOT_NXT	B5	I/O request for data is out of sequence
E_AVAIL_CT	B6	I/O request exceed available data count
E_PARITY	B7	Parity error
E_PIF_REV	B8	No PIF revision level supplied
E_TIMER_ID	B9	Timer id error

E_UNDRUN is a warning, which is available in the trace buffer in the DASM Response Block. During a warning, Byte 128 is not updated.

Note that the following has been removed:

E_SRAM 89 Error in Static RAM

Cont.

Error	Hex	Meaning
M_REJECTED	95	Message Was <REJ>ected By Partner
M_BADREPLY	96	Unexpected Reply Was Received
E_NOTREADY	97	Camera Is Not Ready
E_TIMEOUT	98	EXEC Function Did Not Complete In Time
E_TOOLONG	99	Received Telegram Has Too Many Characters
E_UNEXPECTED	9A	Laser Camera Replied With An Unexpected Telegram
E_BADPARAM	9B	Bad Parameter From Host Found In CMDBLK

— **VRTX Error Codes** —

E_VRTX	C1-F4	<i>VRTX Error Range</i>
	C1	Task ID Invalid
	C2	No Task Control Block Available
	C3	No Memory Available
	C4	No Memory Block

C5	Mailbox in Use
C6	Message of Zero
C7	Buffer Full
C8	WAITC is in Progress
C9	Invalid System Call
CA	Timeout
CB	No Message Present
CC	Queue ID Error
CD	Queue Full
CE	Partition ID Error
CF	Fatal Initialization Error
D0	No Character Present
D1	Invalid Configuration Parameter at Init
D2	Invalid Parameter to PCREATE/PEXTEND
E0	No Component Vector Table
E1	Invalid Component
E2	Invalid Opcode For Component
F0	No Control Block Available
F1	Event Flag Group or Semaphore ID Error
F2	Tasks Pending on Event Flag Group or Semaphore
F3	Event Flag Group or Semaphore is Deleted
F4	Event Flag Group Already Set or Overflow

Appendix C

DASM II VDB Diagrams

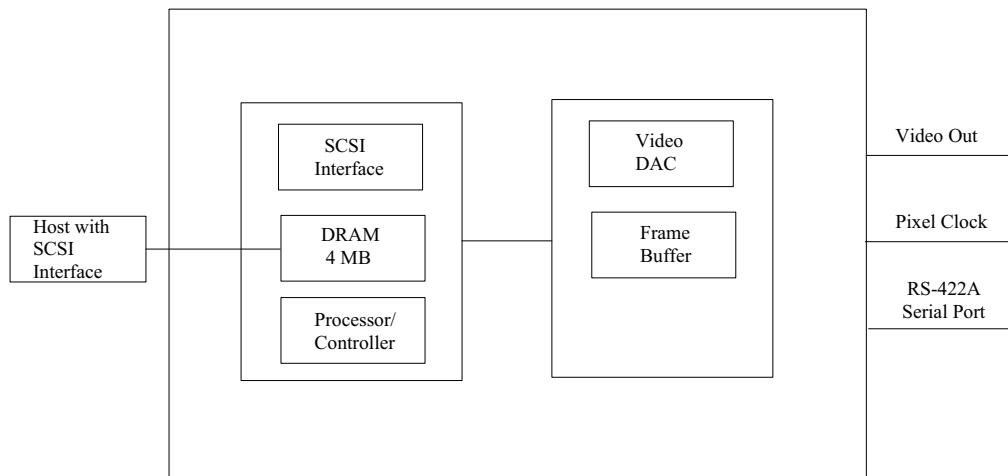
C.1 INTRODUCTION

This section contains the following diagrams for the DASM II VDB:

- ❑ DASM II VDB System Block Diagram
- ❑ DASM Controller Block Diagram
- ❑ VDB Block Diagram

C.2 DASM II VDB SYSTEM BLOCK DIAGRAM

Figure C-1: DASM II VDB System Block Diagram



The DASM II VDB System Block diagram consists of five major elements, as follows:

- ❑ SCSI Interface
- ❑ DRAM
- ❑ Processor / Controller
- ❑ Video DAC
- ❑ Frame Buffer

C.3 DASM CONTROLLER BLOCK DIAGRAM

Figure C-2 displays the DASM Controller Block Diagram.

The DASM Controller consists of six major blocks, as follows:

- ❑ Processor block
- ❑ EPROM
- ❑ Memory Access Controller
- ❑ SCSI Controller
- ❑ Common Data Memory
- ❑ Personality interface

C.3.1 Processor (PROC)

The PROC consists of a MC68001 running at 10MHz, with its own local Static RAM and its own program memory EPROM. The PROC also includes a 68901 multi-function peripheral chip, which handles interrupts and serial communications.

C.3.2 EPROM

The EPROM contains the software to operate the DASM II VDB. The EPROM is connected directly to the 68001 via the CPU Address and Data buses.

C.3.3 Memory Access Controller (MAC)

The Common Data Memory is accessed by three different sources: the SCSI DMA, the PIF DMA, and the 68001. The MAC arbitrates the

Common Bus. The SCSI DMA and PIF DMA logic are implemented in the MAC. The PROC accesses the DMEM via its own address bus.

The SCSI DMA consists of 24-bit address counters. It transfers data between the SCSI bus and DMEM when allowed by the Common Bus arbitrator. The SCSI DMA releases the Common Bus (COM) if a request is pending and 8-bit data was transferred.

The PIF DMA consists of 24-bit address counters, and 16-bit word counters. It transfers data in burst mode (8 bytes before releasing the COM). The PIF DMA can operate in automatic chaining operation. At the end of a DMA, a new DMA immediately starts without any interaction with the microprocessor. In chaining mode, the new address and word counter are loaded in temporary registers by the microprocessor during the current DMA.

The COM bus arbitrator gives the highest priority to a refresh cycle. When no refresh request is pending, a "round robin" arbitration scheme is implemented. The DMAs release the COM, only if another request is pending.

C.3.4 SCSI Controller (SCSI)

The SCSI controller is a WD33C93A controller. It includes bi-directional data latches to increase the transfer rate between the SCSI and the Common Memory Data.

C.3.5 Common Data Memory (DMEM)

The Common Memory Data bus is 8-bits wide and consists of 4 Mbytes of Extended Data Out Dynamic Random Access Memory. One Kbyte is reserved for the Command Block (Block 0) and the Response Block (Block 1). All other DRAM is available for the user's data.

C.3.6 Personality Interface (PIF)

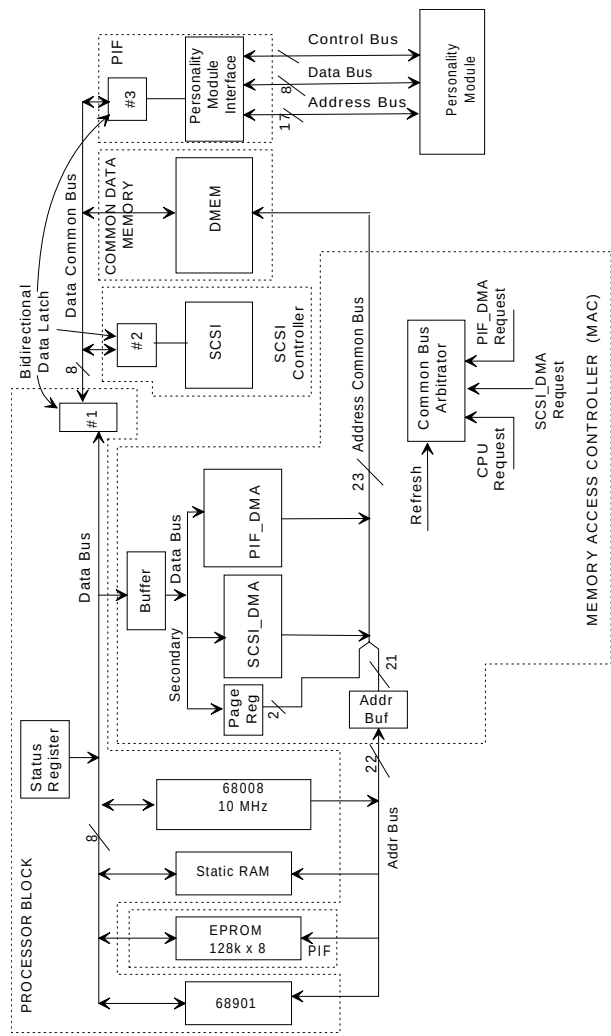
The PIF acts as a slave device, under the control of the 68001 microprocessor, for such operations as writing the Mode Register or reading the Status Register.

During DMA cycles the Memory Access Controller manages PIF activity. Data from the Common Memory is first latched, to permit a higher DMA transfer rate, then enabled on the 8-bit PIF Data bus.

The connections consist of five main groups of signals, as described below.

- ❑ Address lines (17 low addresses)
- ❑ Data Bus (8-bit) from 68001
- ❑ Data Common Bus (8-bit) from (DMEM)
- ❑ DMA handshake signals
- ❑ Serial lines

Figure C-2: DASM Controller Block Diagram



C.4 VDB CIRCUITRY

Figure C-3 displays the VDB circuitry.

This VDB circuitry contains the following components:

- ❑ CPU Interface
- ❑ DMA Interface
- ❑ Field Memory
- ❑ Graphics Controller
- ❑ Video DAC and Sync Summer

C.4.1 CPU Interface

The DASM can configure various VDB parameters through a set of control registers. Included in the CPU Interface are the DASM serial communications signals, TXD and RXD. This two-wire link is converted to differential RS-422 levels (RS-485 compatible), and brought out to a 25-pin D-connector on the front panel.

C.4.2 DMA Interface

The DASM Controller has an independent 10 Mbytes/sec. DMA port. On the PIF side, this port is buffered and connected to the Shared Memory. This mechanism allows the DASM to transfer image data directly into the Field Memory at high speeds, without incurring any CPU overhead.

C.4.3 Field Memory

The Field Memory consists of an OKI MSM518222. An image is transferred from DMEM, Common Data Memory, over the Personality Interface Bus and stored in Field Memory. Due to its self-refreshing feature, this device can be viewed as a 256K x 8-bit FIFO.

C.4.4 Graphics Controller

The Graphics Controller performs the following:

1. Produces the video signals Composite Sync and Blank
2. Controls the Field Memory Read Operation
3. Generates the 16-Level Grey Scale
4. Multiplexes the 16-Level Grey Scale, Image Data and 12-Line Border

C.4.4.1 16-Level Grey Scale Counter

A 32 pixels x 16 level grey scale bar resides to the left side of image field. The Grey Scale Step Size decrements by 17 every 32 lines, beginning with white (255) at upper left corner. Conveniently 255 translates to 0xFF hex, therefore, decrementing a four bit counter once every 32 lines, and using the result as the upper and lower nibble has the desired effect. For example; FF, EE, DD etc.

C.4.4.2 Image Data and the Field Memory

Image data is read from the Field Memory when the Horizontal and Vertical Counters are both within the image field range. In addition, to prime the pipeline one pixel is read prior to entering into the image field.

C.4.4.3 Pixel Multiplexer

The Pixel Multiplexer directs the data to the Video Digital to Analog Converter (DAC) from three sources:

Serial Data:	Image Data from the Field Memory
Grey Scale Data:	The current value in the Grey Scale Register
Border Data:	The value stored in the Border Register

The Pixel Multiplexer simply decodes the Vertical and Horizontal Count for each data type and stores the appropriate data in the Pixel Data Register on the rising edge of pixclk. Also, two control signals GreySel1 and GreySel0 allow the user to turn off, display 0 data, in either the image field or the Grey Scale area respectively.

C.4.5 Video DAC and Sync Summer

C.4.5.1 Video DAC

The DASM II VDB uses the 10-bit Harris HI2315 Video Digital to Analog Converter. The HI2315 was selected because of its low glitch energy specification of 30 picovolt/seconds maximum for any step size.

The VDB only requires 8-bits of resolution, thus the least significant bit is pulled high and the next lowest bit is held low. This is an attempt to reduce the spike noise, associated with glitch energy, even further by never allowing an even code to be converted. This will result in a 4 mV DC offset.

The HI2315 provides a current output, which is configured to invert in our application.

The DAC is updated by the pixel clock at a rate of 24.194 MHz.

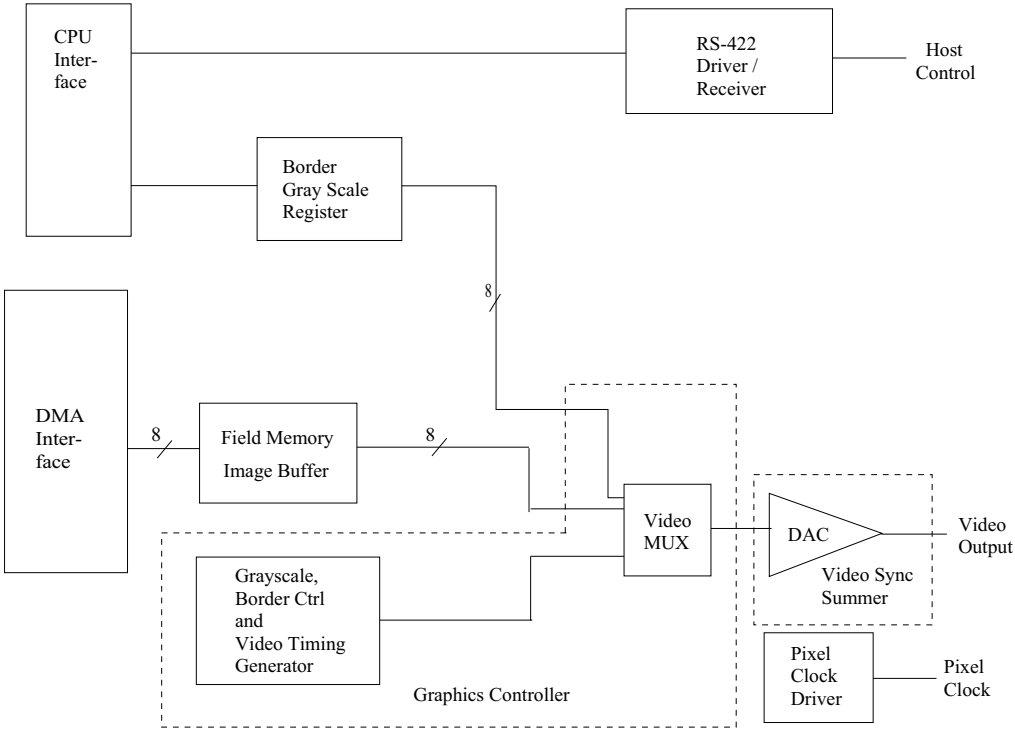
C.4.5.2 Sync Summer Circuit

The HI2315 provides a pin for blank, unfortunately, the blank only outputs a black level when active. Thus addition circuitry is needed to inject Blank and Sync onto the Video signal, this circuit is called the Sync Summer. The Sync Summer is basically a two-bit DAC, with one bit weighted for Sync and the other for Blank. This circuit employs a modified Wilson Current Source controlled by two n-Channel FETs. The gates of two 2N7002 are tied to the Blanking and Sync signals.

When active, current is sourced into the summing node. The summing node is the point that both the Inverted DAC output and the Current Source is connected. In this case at the input of the divider network. A DC offset of .071V and .286V are developed across a load resistor for Blank and Sync levels respectively.

The Sync Summer circuit also contains an HI1114 850MHz buffer capable of driving a 75 Ohm Coax Cable. The HI1114 also inverts the video signal produced by the DAC.

Figure C-3: VDB Block Diagram



Appendix D

Specifications

D.1 MECHANICAL

The DASM II VDB is entirely contained within a single metal enclosure. This enclosure is approximately 10"W x 2.5"H x 10L" and sits on four 1/8" hard rubber feet to prevent scratching of the support surface. So called "human interfaces" located on rear panel are; a power ON/OFF switch and two 50 pin Centronics type SCSI connectors (SCSI IN and SCSI OUT) with latching cable restraints. The SCSI ID selection rotary switch and SCSI terminator disable switch are accessible on the bottom of the chassis. The front Panel of the DASM II VDB contains a 25-pin female "D" connector, six green LEDs, and two standard BNC connectors (PIXEL CLOCK, and VIDEO).

D.2 ENVIRONMENTAL

D.2.1 Environmental Conditions

Operating

Operating temperature limits from 0 Centigrade to 40 Centigrade

Operating relative humidity limits from 20% to 93% non-condensing

Operating altitude limits from sea level to 10,000 feet above sea level

Non-Operating

Non-Operating and Storage

Non-Operating temperature limits from -25 Centigrade to 70 Centigrade

Non-Operating relative humidity limits from 20% to 95% non-condensing

D.3 SCSI INTERFACE

SCSI Interface:

8-bit SCSI-1, Command Set supported.

SCSI-1 operation is compatible with a SCSI-2 host.

SCSI Transfer Rates:

Synchronous Mode: 4.0 Mbytes/sec Max

Asynchronous Mode: 2.5 Mbytes/sec Max

Note: These rates are 'theoretical maximums'. Actual rates will vary. A typical SUN SPARCstation-1 transfers data from DASM memory to SUN memory @ 1.5 Mbytes/sec.

SCSI Connector:

The following SCSI connector Pin Out table describes the signal associated with each pin in the SCSI cable. All odd pins except pin 25 must be connected to ground. Pin 25 must be left open. Some earlier SCSI devices also may connect this pin to ground. Also note that a minus sign (-) next to the signal indicates active low.

Signal	Pin Number	Signal	Pin Number
-DB(0)	2	GROUND	28
-DB(1)	4	GROUND	30
-DB(2)	6	-ATN	32
-DB(3)	8	GROUND	34
-DB(4)	10	-BSY	36
-DB(5)	12	-ACK	38
-DB(6)	14	-RST	40
-DB(7)	16	-MSG	42
-DB(P)	18	-SEL	44
GROUND	20	-C/D	46
GROUND	22	-REQ	48
GROUND	24	-I/O	50
TERMPWR	26		

Note: "-" means active low.

D.4 SERIAL LINE INTERFACE

EIA Standard RS-422A, RS-485 compatible 6-wire interface on a 25-pin female D-connector.

25 PIN SERIAL LINE CONNECTOR

Signal	Pin	Function
RX (+)	8	Serial Input to DASM (+ diff)
RX (-)	21	Serial Input to DASM (- diff)
TX (+)	9	Serial Output from DASM (+diff)
TX (-)	22	Serial Output from DASM (- diff)
SIGGND	7	DC (logic) Ground.
CASE GND	1	Earth Ground. Shunt selectable.

Communications Parameters:

	Software Selectable	Default Values
Baud Rate:	standard rates up to 19.2K	1200 baud
Word Length:	5,6,7,8 data bits	8 bit
Start/Stop Bits:	0/0, 1/1, 1/1.5, 1/2	1/1
Parity:	Even, Odd, No Parity	Even
Format:	Synchronous or Asynchronous	Asynchronous

D.5 INPUT

AC Power Requirements

Input range: 100-240VAC

Frequency: 50-60Hz

Current: 0.6A

AC Power Input Connector

The DASM II VDB provides an industry-standard 3-pin power entry IEC connector for connection to the AC Line.

DC Power Requirements

5.0VDC @ 1.2 Amps Typical, 1.5 Amps Maximum

12.0VDC @ 0.125 Amp Typical, 0.14 Amp Maximum

Combined noise and ripple must be less than 1% of the supplied DC Voltage.

CAUTION: To avoid damaging the video DAC, both 5V DC and 12V DC must be supplied.

DC Power Mating Connector

Housing part number AMP Incorporated 1-480424-0

Pin part number AMP Incorporated 60619-1

Pin assignment:

J2-4 +5.0VDC

J2-3 Return

J2-2 Return

J2-1 +12VDC

SCSI Bus

SCSI-1 single-ended.

Serial Port

RS-422A/RS-485

D.6 OUTPUTS

Video Output (measured into 75 ohms at the BNC connectors)

Amplitude: 1 Volt peak-peak

Video: 0.643V +/- 10%

Setup: 0.071V +/- 10%

Sync: 0.286V +/- 10%

DAC Resolution: 8 bits

Differential linearity: +/- 1 LSB Max

Glitch Area: 80 picovolt-seconds Max. for any step size

Rise and Fall Times: < 10 nsec, 10%-90%

FS Setting Time: 7.5 nsec typical to 1 LSB

Transfer Function: Guaranteed Monotonic

Noise Level: < 5.0 mV peak-peak, synchronous and non-synchronous noise combined.

D.C. Offset: +/- Vdc referred to ground.

Pixel Clock Output (measured into 75 ohms at the BNC connectors)

Logic Family: F series, TTL

Output Low Level: 0.8 Vdc Max

Output High Level: 2.0 Vdc Min

Output Period: 41.336 nanoseconds, +/- .01%

Transition Times: 10 nsec Max, 10%-90%

D.7 FUNCTIONAL GROUNDING

Safety Ground

The DASM II VDB is provided with a 3-pin AC Power Cord. This power cord provides the Safety Ground between the exposed metal parts of the DASM II VDB and the local electrical ground.

Serial Line Interface Ground

The RS-422 Serial Line connector provides two internal ground signals. One (pin 7) is a DC (logic) ground and is connected to the DASM II VDB ground plane. The second ground (pin 1) is connected to a 3-pin jumper field. This arrangement provides the option of 'floating' pin 1 (no connection to the DASM II VDB chassis ground) or optionally connecting pin 1 of the serial line to the DASM II VDB chassis ground.

The DASM II VDB will be shipped with pin 1 floating, unless otherwise specified.

Coax Cable Ground

Two 75 Ohm BNC connectors provide the VIDEO and PIXEL CLOCK outputs. These BNC connectors provide capacitive decoupling between the connector body and the DASM II VDB enclosure. EMI on the coaxial shield is shunted to the chassis ground through the capacitor.

D.8 CHARACTERISTICS

DASM Data Memory

Size: 4 Mbytes only
 Usage: First 1 Kbyte reserved for system use.
 (Block 0 and Block 1).
 All other DRAM is available for data.

<u>Video Timing Specifications</u>	<u>60-Hz</u>	<u>50-Hz</u>
Pixel Freq.	24.192 MHz	24.192 MHz
Pixel Period	41.336 nsec	41.336 nsec
Horizontal Line Freq.	33.6 kHz	33.6 kHz
Horizontal Line Width	720 pixels	720 pixels
Horizontal Active	544 pixels	544 pixels
Horizontal Blanking	176 pixels	176 pixels
Horizontal Front Porch	26 pixels	26 pixels

Horizontal Sync	76 pixels	76 pixels
Horizontal Back Porch	74 pixels	74 pixels
Vertical Frame Freq.	60 Hz	50 Hz
Vertical Frame Time	560 lines	672 lines
Vertical Active	524 lines	524 lines
Vertical Blanking	36 lines	148 lines
Vertical Sync	3 lines	3 lines
Vertical Back Porch	30 lines	86 lines
Vertical Front Porch	3 lines	59 lines
Scanning Format	Non-Interlaced	Non-Interlaced

Video Display Format

Visible Field:	544 pixels x 524 lines
Image Field:	512 pixels x 512 lines
Grey Scale field:	32 pixels x 16 level gray scale bar on left side of image field
Grayscale:	software-selectable, ON/OFF
Grayscale-OFF value:	0 (black)
Initial Grey Scale:	255 (white) at upper left corner
Grey Scale Step Size:	Decrement by 17 every 32 lines
Border Field:	12 lines at bottom of Visible Field
Border Field Value:	Any 8-bit value, software programmable

Physical Characteristics

Standalone enclosure with power supply: 10"W x 2.5"H x 10"D

Front Panel: 4 - green status LEDs for DASM Controller
2 - green status LEDs for VDB
2 - BNC connectors (Composite Video, Pixel Clock)
1 - 25-pin female D-connector for RS422 serial port

Rear Panel: 1 - 50-pin "Centronics" female connector: SCSI IN
1 - 50-pin "Centronics" female connector: SCSI OUT
1 - VAC Power Entry IEC connector
1 - Power Switch

Bottom: 1 - SCSI ID Switch
1 - Active Termination Switch

Cables

AC Power Cord – 6 foot minimum

SCSI Cable

Length:	3 feet
Connectors:	1 – SCSI-1 50-pin, Centronics-type 1 – SCSI-2 50-pin, subminiature-type
Compatible Hosts:	SUN, Hewlett-Packard, IBMRS 6000, others.

External SCSI Terminator Plug

Type:	SCSI-1 Single Port, Centronics-type
Mode:	Single-Ended Passive or Active

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